ST. PAUL'S SR. SEC. SCHOOL, MATHURA

HOLIDAY HOMEWORK 2026-27

CLASS - X

Maths

ST. PAUL'S SR. SEC. SCHOOL , MATHURA
Holiday Homework
Class 10 Maths

Real Numbers

1. A teacher distributes 96 chocolates and 144 candies equally among students without any item left. What is the greatest number of students possible?

2. Use prime factorization to find the HCF of 867 and 255.

3. Prove that $\sqrt{7}$ is an irrational number.

4. Find the LCM and HCF of 72 and 120. Verify: LCM × HCF = Product of the numbers.

5. Two alarm clocks ring after every 18 minutes and 24 minutes respectively. If they ring together at 8:00 AM, after how much time will they ring together again?

6. Express 0.625 as a rational number in simplest form.

7. Without actual division, determine whether the decimal expansion of 77/210 is terminating or non-terminating recurring.

Polynomials

8. Find the zeroes of the polynomial: $x^2 - 5x + 6$

9. Verify the relationship between zeroes and coefficients for: $2x^2 - 7x + 3$

10. Form a quadratic polynomial whose zeroes are 4 and -3.

11. Divide: $2x^3 + 5x^2 - 4x + 3$ by $(x + 2)$

12. If one zero of the polynomial $x^2 - 7x + 12$ is 3, find the other zero.

13. Find the remainder when $3x^3 - 2x^2 + 4x - 5$ is divided by $x - 2$.

14. Determine whether $(x+1)$ is a factor of: $x^3 + 2x^2 - x - 2$

15. The sum of zeroes of a quadratic polynomial is 8 and their product is 12. Form the polynomial.

16. Find the value of k if $(x-2)$ is a factor of: $x^2 + kx - 6$

Probability

17. A coin is tossed once. Find the probability of getting: (i) at least 1 head (ii) atmost 1 tail

18. A die is rolled once. Find the probability of getting: (i) a prime number (ii) a number greater than 4

19. A bag contains 5 red balls, 3 blue balls and 2 green balls. One ball is drawn at random. Find the probability that it is: (i) blue (ii) not green

20. Cards numbered 1 to 20 are mixed well. One card is drawn at random. Find the probability that the number drawn is: (i) divisible by 5 (ii) an even number

ACTIVITY 1: To Find Two real zeroes of the quadratic polynomial $x^2 - 4x + 3$ graphically and also verify the relationship.

ACTIVITY 2: To verify graphically that the pair of linear equation x+y-10=0 ,x+y+4=0 in which $\frac{a1}{a2} = \frac{b1}{b2} \neq \frac{c1}{c2}$ give a parallel straight line.